

Topics

- Density
- Openness is Relative
- Compactness definition + examples
- Compactness & intrinsic

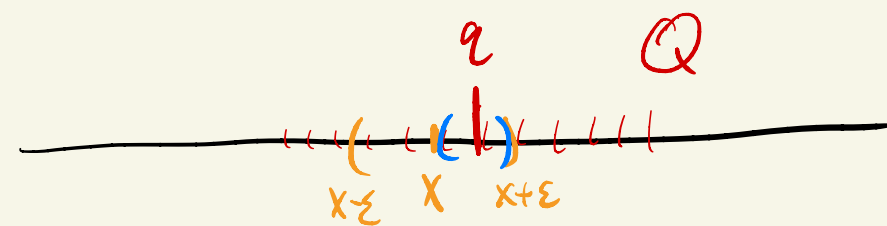
Def $E \subset X$ is dense if every $x \in X$ is either in E or a limit point of E . That is, $E \cup E' = X$.

Ex

① X is dense in X for any metric space X .

② $(0,1)$ is dense in $[0,1]$.

③ $\mathbb{Q}$ is dense in $\mathbb{R}$.



proof of 3: Let $x \in \mathbb{R}$, and let $\varepsilon > 0$. There exists

some $q \in \mathbb{Q}$ s.t., $x < q < x + \varepsilon$ by our old "density of $\mathbb{Q} \subset \mathbb{R}$ " theorem.

This implies $q \in N_\varepsilon(x) \setminus \{x\}$, so x is limit point of $\mathbb{Q}$.

Def Let $E \subset X$ be a subset of a metric space.

• $\bar{E} := E \cup E'$ is the closure of E .

Examples

• $\overline{(a, b)} = [a, b]$

• $\overline{\mathbb{Q}} = \mathbb{R}$

• $\overline{\left\{\frac{1}{n} \mid n \in \mathbb{N}\right\}} = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\} \cup \{0\}$

Theorem (2.27) " $\bar{E}$ is the smallest closed set containing E "

(a) $\bar{E}$ is closed

(b) $E = \bar{E}$ iff E is closed

(c) $\bar{E} \subset F$ for every closed $F \subset X$ s.t. $E \subset F$

Proof

(a) $x \notin E \cup E' \Rightarrow x$ has a nbhd N s.t. $N \cap E = \emptyset$.

$\Rightarrow N \cap E'$ is also empty because every $y \in N$ has a nbhd $\tilde{N} \subset N$ which won't intersect E .

$\Rightarrow \bar{E}^c$ is open

$\Rightarrow \bar{E}$ is closed

(b) E closed $\Rightarrow E' \subset E$, so $E = E \cup E'$.

$E = \bar{E} \Rightarrow E$ closed by (a)

(c) Suppose F closed, and $F \supset E$.

F closed $\Rightarrow F \supset F'$

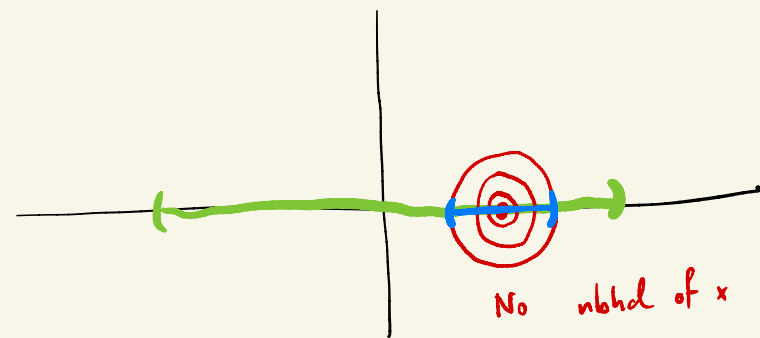
$F \supset E \Rightarrow F' \supset E'$

$F \supset E$ and $F \supset E' \Rightarrow F \supset \bar{E}$.

Relative openness

Def $E \subset Y \subset X$ is open "in Y " or "relative to Y " iff it is open as a subset of the metric space Y : $\forall p \in E, \exists r > 0$ s.t. $q \in Y$ and $d(p, q) < r \Rightarrow q \in E$.

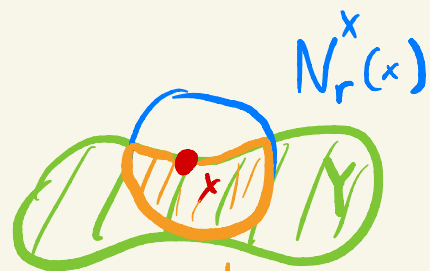
Example: $(-1, 1) \times \{0\}$ is open in $\mathbb{R} \times \{0\}$, but not in $\mathbb{R}^2$.



No nbhd of x in $\mathbb{R}^2$ contained in $(-1, 1) \times \{0\}$

But, there is a nbhd in $\mathbb{R} \times \{0\}$ contained in $(-1, 1) \times \{0\}$.

Def Let $N_r^Y(p) = \{q \in Y \mid d(p, q) < r\}$



$$N_r^Y(x) = N_r(x) \cap Y$$

Theorem (2.30) Let $Y \subset X$. $E \subset Y$ is open in Y iff $E = Y \cap G$ for some open $G \subset X$.

proof:

$\Leftarrow$ Suppose G is open in X . Let $p \in G \cap Y$.

$N_r(p) \subset G$ for some $r > 0$, implying $N_r(p) \cap Y \subset G \cap Y$,
so $G \cap Y$ is open in Y .

$\Rightarrow$ Suppose E open in Y .

for each $p \in E$ $\exists r_p$ s.t. $N_{r_p}^Y(p) \subset E$.

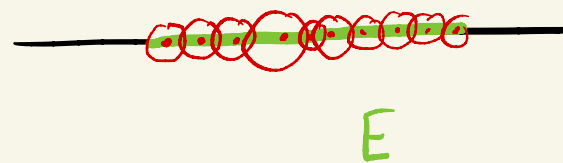
Let $G = \bigcup_{p \in E} N_{r_p}^X(p)$

$G \cap Y = \left(\bigcup_{p \in E} N_{r_p}^X(p) \right) \cap Y = \bigcup_{p \in E} N_{r_p}^Y(p)$ by distributivity.

$E \subset Y \cap G$ because each p covered by $N_{r_p}^Y(p)$.

$E \supset Y \cap G$ because $N_{r_p}^Y(p) \subset E$ for all p .

$\Rightarrow E = Y \cap G$.

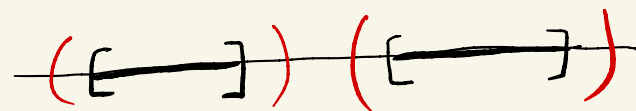


Corrolary, $E \subset Y$ is closed in Y iff $E = Y \cap G$
for some closed set $G \subset X$.

proof idea: Play with complements.

Examples

• $[0,1], [2,3]$ are open in $[0,1] \cup [2,3]$



• $(a,b) \cap \mathbb{Q}$ is open in $\mathbb{Q}$

Def A metric space X is connected if there do not
exist disjoint, non-empty, open $\wedge U, V \subset X$ s.t. $X = U \cup V$.
(equiv. closed, or clopen)

Examples

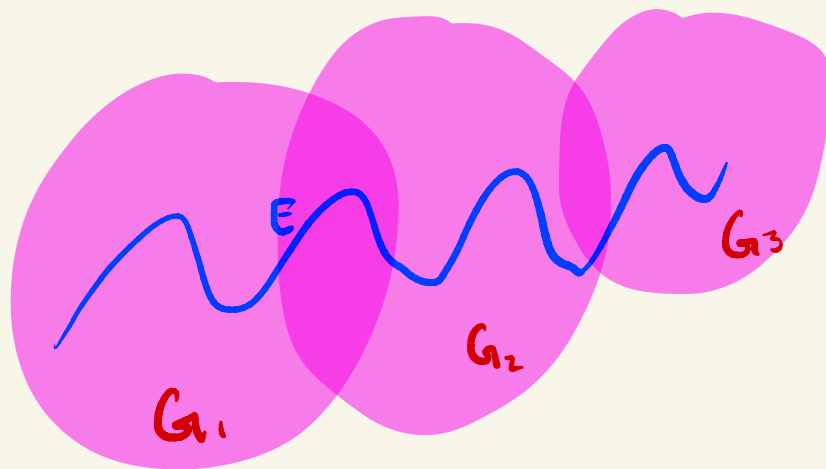
• $\mathbb{R}, [a,b], (a,b)$ connected. We will come back to prove this.

• $[0,1] \cup [2,3]$ disconnected.

• $\mathbb{Q} = \{q \in \mathbb{Q} \mid q < \sqrt{2}\} \cup \{q \in \mathbb{Q} \mid q > \sqrt{2}\}$ disconnected.

Compactness

Def An open cover of $E \subset X$ is a collection $G_\alpha \subset X$ of open sets s.t. $E \subset \bigcup_\alpha G_\alpha$.



Def $E \subset X$ is compact if every open cover G_α has a finite subcover: $\exists \alpha_1, \dots, \alpha_k$ s.t. $E \subset G_{\alpha_1} \cup \dots \cup G_{\alpha_k}$.

Are the following subsets of $\mathbb{R}$ compact?

- $\{1, 2, 3\}$
- $\mathbb{N}$
- $\{\frac{1}{n} \mid n \in \mathbb{N}\}$
- $\{\frac{1}{n} \mid n \in \mathbb{N}\} \cup \{0\}$

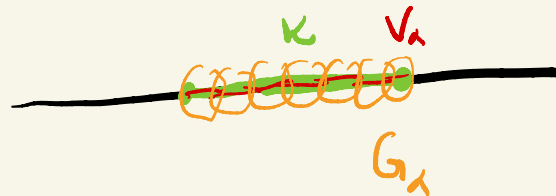
Theorem (2.33) Suppose $K \subset Y \subset X$. K is compact in Y
 iff K is compact in X .

Meaning: Compactness is intrinsic, K is a compact metric space.

(Just follow your nose using theorem 2.30)

proof: $\checkmark$ Suppose K cpt. in X .

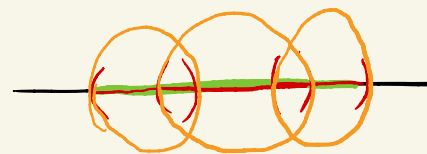
Let $\{V_\alpha\}$ be an open cover of K in Y .



There are G_α open in X s.t. $G_\alpha \cap Y = V_\alpha$.

Thus there is finite subcover $G_{\alpha_1}, \dots, G_{\alpha_n}$ of K in X .

$V_{\alpha_1}, \dots, V_{\alpha_n}$ is a finite subcover in Y .



Suppose K cpt. in Y .

Let $\{G_\alpha\}$ be an open cover of K in X .

Let $V_\alpha = G_\alpha \cap Y$. $\{V_\alpha\}$ is an open cover of K in Y .

Let $V_{\alpha_1}, \dots, V_{\alpha_n}$ be a finite subcover in Y .

$G_{\alpha_1}, \dots, G_{\alpha_n}$ is a finite subcover in X .